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SECP3213 - BUSINESS INTELLIGENCE

SECTION 01

PROJECT PHASE 1: INTRODUCTION (PROPOSAL STAGE)

PROJECT TITLE:

E-COMMERCE SALES AND MARKET ANALYSIS ON AMAZON INDIA

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Problem Background

This section examines the current problem, providing the reason to show how important this problem that is worth to be analyzed into actionable business insights.

Business Context

In recent years, the e-commerce industry keeps evolving from a supplementary retail channel into the primary commerce engine. In emerging markets, this growth has been especially pronounced, with platforms like Amazon capturing a significant share of both domestic and international retail activity. Understanding the dynamics of this growth has become a strategic priority for businesses seeking to remain competitive in an increasingly data-driven marketplace. This project focuses on a real-world e-commerce sales dataset sourced from Kaggle. Including three datasets which are Amazon Sale Report, Sale Report, and International sale Report in csv format.

Importance of Problem

Despite the wide-range of accessible data, many e-commerce businesses struggle to change raw sales data into valuable information. To prevent companies from developing a complete understanding of their performance, most data collection like sales, report, and transaction are stored and analysed in isolation. Thus, it makes it more difficult to answer critical business questions such as most revenue categories or impact on profitability. Furthermore, the dataset reflects the complexity of real-world e-commerce operations without a structured BI solution, businesses are left making pricing, inventory, and logistics decisions based on incomplete or siloed information. This leads to missed revenue opportunities, inefficient stock management, and an inability to capitalize on high-performing markets or channels.

Expected Impact

By developing a comprehensive BI solution that integrates all three datasets using Alteryx for data preparation and Power BI for visualisation, this project aims to deliver the following outcomes:

- **Enable cross-dataset business intelligence** by linking transactional order data, product-level inventory and pricing data, and international sales records into a unified analytical model.
- **Identify high-performing product categories and platforms** to guide pricing strategies, promotional planning, and inventory allocation decisions.
- **Evaluate fulfilment efficiency** by comparing Amazon-fulfilled versus seller-fulfilled orders in terms of volume, revenue, and order status outcomes.
- **Uncover international market opportunities** by analysing cross-border sales patterns, currency distributions, and geographic demand trends.
- **Support data-driven decision-making** at both the operational and strategic level, providing stakeholders with clear, visual dashboards that surface actionable insights from complex, multi-source data.

Ultimately, this BI solution will demonstrate how integrating multiple structured datasets from a single e-commerce ecosystem can produce far richer and more actionable insights than any individual file alone.

Objectives

1. To analyse sales performance across product categories and fulfilment channels to identify top-revenue and high-loss segments.
2. To examine order status patterns (cancelled, returned, shipped) and their impact on overall revenue and operational cost.
3. To compare pricing strategies across multiple platforms and evaluate their effect on profit margins.

Datasets Description

Although all datasets originate from a single Kaggle source, each CSV file represents a distinct data entity which are transactional, inventory/pricing, and financial performance. All of these datasets will be integrated using Alteryx to form a unified BI solution.

Dataset 1 - Amazon Sale Report.csv

This dataset provides detailed insights into Amazon sales data, including SKU Code, Design Number, Stock, Category, Size and Color, to help optimize product profitability.

Details	Description
Source	Kaggle - E-Commerce Sales Dataset
Link	https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data/data?select=Amazon+Sale+Report.csv
Type of data	Structured
Format	.CSV
Size	~128,975 rows × 17 columns

Key Variables

Column Name	Data Type	Description
Order ID	String	Unique identifier for each transaction
Date	Date	Date of sale
Status	Categorical	Order status (Shipped, Cancelled, Pending, Returned)
Sales Channel	Categorical	Platform used (Amazon.in)
Category	Categorical	Product category
Size	Categorical	Product size (S, M, L, XL, XXL, etc.)
SKU	String	Stock Keeping Unit code
ASIN	String	Amazon Standard Identification Number
Courier Status	Categorical	Shipping courier status
Qty	Integer	Quantity ordered

Currency	Categorical	Transaction currency (INR)
Amount	Float	Amount paid by customer
Ship-City	String	Destination city
Ship-State	String	Destination state
Fulfilled-by	Categorical	Easy Ship or other fulfilment method

Dataset 2 - Sale Report.csv

This dataset provides detailed information on e-commerce sales, including SKU Code, Design Number, Stock, Category, Size and Color, to help optimize product profitability.

Details	Description
Source	Kaggle - E-Commerce Sales Dataset
Link	https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data/data?select=Sale+Report.csv
Type of data	Structured
Format	.CSV
Size	~9270 rows × 7 columns

Key Variables

Column Name	Data Type	Description
SKU Code	String	Unique identifier for each product
Design No.	String	Unique identifier for each design
Stock	Integer	Number of items in stock
Category	String	Type of product
Size	String	Size of the product
Color	String	Color of the product

Dataset 3 - International Sale Report.csv

This file provides detailed information on international sales, including customer, style, SKU, size, pieces, rate, and gross amount.

Details	Description
Source	Kaggle - E-Commerce Sales Dataset
Link	https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data/data?select=International+sale+Report.csv
Type of data	Structured
Format	.CSV
Size	~37,400 rows × 9 columns

Key Variables

Column Name	Data Type	Description
Style	String	Style of the product
SKU	String	Stock Keeping Unit
Size	String	Size of the product
DATE	Date	Date of the sale
Months	String	Months of the sale
CUSTOMER	String	Name of the customer
PCS	Integer	Number of pieces/quantity ordered internationally
RATE	Float	Price per piece
GROSS AMT	Float	Total gross revenue from the international order

Proposed BI Approach

To develop a robust Business Intelligence solution, our group will implement an end-to-end data pipeline using **Alteryx Designer** for the ETL which is extract, transform and load process and **Power BI** for the final visualization and insight delivery.

Tool Utilization

- **Alteryx Designer**

We will use alteryx designer as a main platform for the processing and integrating data. We will use it to ingest the three separate csv files, resolve the data consistencies and also perform the complex joins that are difficult to handle manually.

- **Power BI**

Power BI will be used to transform the data into a dashboard that can be easily understood by the stakeholders. This will help them to get an insight for the decision making process.

Data Integration, Cleaning, and Transformation

The data preparation phase in Alteryx will follow a structured workflow:

- **Data Integration:**

We will use the join tool to combine the Amazon Sale Report with the Sale Report. This will allow us to analyze the sales performance against current stock levels. Other than that, we will also integrate the International Sales Report to provide a view about the global vs domestic revenue.

- **Data Cleaning**

- Handling nulls by using the data cleansing tools to address the missing value
- Filtering by applying the filter tool to separate "Cancelled" or "Returned" orders from successful transactions to calculate "Actual Revenue"
- Data Type Correction where we use the Select Tool to ensure "Date" fields are in the correct ISO format and that "Amount" or "Rate" are treated as fixed decimals.

- **Data Transformation**

- Aggregation where we will use summarize tools to group sales by month, category and region.
- Feature Creation where we will use Formula Tools to calculate essential KPIs
- Currency standardization where we will unified the currency for the comparison.

Planned Analysis and visual Forms

Our analysis will focus on identifying revenue drivers and operational bottlenecks. We plan to conduct the following:

1. **Sales Trend Analysis:** To identify seasonal peaks and troughs in online shopping behavior.
2. **Product Category Performance:** To pinpoint which categories like Western Wear vs. Ethnic Wear that generates the highest volume versus highest profit.
3. **Inventory Health Check:** Comparing stock levels against sales velocity to identify overstocked or understocked items.
4. **Geographic Distribution:** Mapping where the majority of domestic orders are fulfilled.
5. **Order Fulfillment Efficiency:** Analyzing the ratio of shipped vs. cancelled/returned orders.

Proposed Visual Forms

1. **Line Chart:** To visualize sales and revenue trends over time (monthly/daily).
2. **Stacked Bar Chart:** To compare revenue across different product categories and sizes.
3. **Symbol Map:** To show the density of orders across different cities and states.
4. **Donut Chart:** To represent the distribution of order statuses like shipped, cancelled, or returned.
5. **Scatter Plot:** To analyze the correlation between "Stock Levels" and "Quantity Sold" to identify high-performing SKUs.